

Official AI Disclosure Statement

Samsung PRISM GenAI Hackathon 2026–27 (3rd Edition) • Theme 4: Streaming Live RAG

Track / Theme: Theme 4 — Streaming Live RAG	Submission Tag: PRISM_GENAI_HACKATHON_Y2026
Team Name: VIT_RAGnarok	Target Commit: 4d0cba8
Institution: Vellore Institute of Technology (VIT)	Date: September 30, 2026
Repository: https://github.com/codewithyuvi/Streaming-Live-Rag	Demo Video: Google Drive Submission Link

1. Executive Summary & Policy Adherence

This document serves as the formal AI Disclosure Statement for Team VIT_RAGnarok participating in the Samsung PRISM GenAI Hackathon 2026–27 (Theme 4 – Streaming Live RAG). In strict compliance with hackathon guidelines, academic integrity, and responsible AI engineering principles, this statement provides a transparent accounting of all Generative AI developer tools and assistants utilized, the explicit boundary between human algorithmic authorship and AI-assisted generation, the neural models integrated into the live runtime production architecture, and our affirmation of originality.

2. Team Member Information & Roles

Member Name	Registered Email	Program & Year	Primary Responsibilities
Yash	Yash.2024a@vitstudent.ac.in	B.Tech CSE (Pre-Final Year)	System Architecture, 2-Stage Speculative Controller, Async Pipeline, Packaging
Yuvraj Bansal	yuvraj.bansal2024@vitstudent.ac.in	B.Tech CSE (Pre-Final Year)	Hybrid Qdrant Search, Dense/Sparse Search, FastEmbed ONNX GPU, Reranker
Aditya Gupta	aditya.gupta2024c@vitstudent.ac.in	B.Tech CSE (Pre-Final Year)	Live VS Arena, Real-time Web Speech & Audio Streaming, Waveform UI
Jaanya Bagdi	Jaanya.bagdi2024@vitstudent.ac.in	B.Tech CSE (Pre-Final Year)	Deterministic Grounding Validator, Ephemeral Session Memory Lineage, Gates G1–G6

3. Inventory of AI Tools Utilized During Development

Tool / Assistant	Provider	Scope & Modality	Application Area &
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			Verification
Google Antigravity & Gemini 2.5	Google DeepMind	Pair-programming agent, code intelligence	Scaffolding async queue boilerplate, ONNX dynamic DLL loading, unit test expansion. Every line inspected and verified.
GitHub Copilot	GitHub / OpenAI	Contextual code completion	Fast typing of boilerplate dataclasses and CSS layouts. Human developer reviewed all completions.
OpenAI ChatGPT (GPT-4o)	OpenAI	Conversational research assistant	Research literature review (Speculative RAG, DAG planning), regex debugging. Zero canned code imported.
Claude Sonnet 5	Anthropic	Technical writing & review	Reviewing edge-case failure modes in async WebSocket reconnection and reviewing documentation drafts.

4. Breakdown of Human Authorship vs. AI Assistance

- Core Architecture & Algorithmic Design (100% Human): Continuous speculative retrieval mid-speech, 2-stage asynchronous controller/synthesizer split, 4-phase honest thought stream, and live VS Arena head-to-head comparison logic were wholly conceptualized and designed by team members.
- Back-End Pipeline Implementation (85% Human / 15% AI-Assisted): Human engineers wrote the core FastAPI routing, WebSocket protocol engine, Qdrant hybrid retrieval scoring, FastEmbed ONNX CUDA provider binding logic, and multi-intent decomposition DAG. AI tools assisted in generating standard type annotations, dataclasses, and error wrappers.
- Grounding Verification & Session Memory (90% Human / 10% AI-Assisted): Formulated the deterministic citation validation algorithm that cross-checks bracketed references ([Doc_ID §Section]) against retrieved chunk provenance, eliminating LLM hallucinations. AI assisted in refining regex pattern boundaries.
- Front-End Dashboard & VS Arena (80% Human / 20% AI-Assisted): Human-designed UI/UX layout, dual-stream WebSocket messaging state machine, and live microphone chunking. AI assisted with CSS layout grid styling and audio visualizer canvas animation helpers.
- Benchmark Harness & Test Suite (85% Human / 15% AI-Assisted): Formulated evaluation Gates G1 through G6. AI assisted in expanding parameterized test matrices for unit test edge cases.
- Documentation & Runbook (85% Human / 15% AI-Assisted): Empirical benchmark figures, latency tables, ADRs, runbooks, and presentation slide copy were drafted from real system runs. AI assisted in formatting and table alignment.

5. Runtime Architecture: Models & Inference Engines

We strictly distinguish between development AI tools and runtime models integrated into the live execution stack:

- Stage 1 Speculative Controller & Decomposer: Groq Cloud LPU running LLaMA 3.3 70B Versatile (sub-300ms intent classification and DAG query decomposition).
- Dense & Sparse Vector Retrieval: FastEmbed ONNX Runtime with NVIDIA CUDA 12 Execution Provider running BAAI/bge-small-en-v1.5 (Dense) and Qdrant BM25 (Sparse lexical) at ~3.0ms per query.
- Neural Reranker: cross-encoder/ms-marco-MiniLM-L-6-v2.
- Stage 2 Streaming Synthesizer: Google Generative AI running Gemini 2.5 Flash for grounded token-by-token streaming synthesis.
- Deterministic Grounding Validator: 100% deterministic code regex context verifier (zero LLM reliance).

6. Ethical Compliance, Originality & Anti-Cheating Attestation

1. Strict Corpus Isolation: The system operates strictly over the designated multi-format corpus (Doc_01 to Doc_06). No external web search or ungrounded generation is permitted.
2. Zero Precomputation or Hardcoded Responses: No answers, queries, thought events, or evaluation outputs are hardcoded. All responses are dynamically generated, retrieved, and validated in real time.
3. Honest Thought Narration: The 4-phase thought stream reflects actual internal pipeline execution timestamps and genuine telemetry. No artificial delays (time.sleep) or fake animations are injected.
4. Fair Head-to-Head Comparison: In the Live VS Arena, both Streaming Live RAG and Naive Sequential RAG operate under identical hardware, model, and network conditions with isolated session states and explicit cache-bypassing (ignore_cache=True).
5. Open Source & License Compliance: All open-source libraries used (FastAPI, Qdrant Client, FastEmbed, ONNX Runtime, Pydantic) are licensed under permissive licenses (MIT, Apache 2.0, BSD-3-Clause).

7. Team Affirmation & Signatures

We, the undersigned team members of VIT_RAGnarok, hereby certify on our honor that all information contained in this AI Disclosure Statement is true, accurate, and complete. We maintain full intellectual ownership over all code and deliverables.

Member Name	Role	Date	Status
Yash	Team Lead	September 30, 2026	Digitally Verified
Yuvraj Bansal	Core Developer	September 30, 2026	Digitally Verified
Aditya Gupta	Core Developer	September 30, 2026	Digitally Verified
Jaanya Bagdi	Core Developer	September 30, 2026	Digitally Verified